Machine Learning Project 1

Chloé Dupuis, Lise Gillin, Adrien Perrini

***Abstract*—This project aims to predict the risk of coronary heart disease using the 2015 Behavioral Risk Factor Surveillance System (BRFSS) dataset. We investigate data preprocessing, feature selection, class imbalance and regression-based classification methods. Our best model, based on ridge regression, achieves an F1 score of 0.415 and an accuracy of 0.835 on AIcrowd.**

I. Introduction

Coronary heart disease (CHD) occurs when blood flow to the heart is reduced by the accumulation of fatty deposits in the coronary arteries. It is one of the leading causes of mortality worldwide, making early detection an important medical challenge.

In this project, we use the 2015 BRFSS dataset to predict whether an individual has experienced coronary heart disease or myocardial infarction (MICHD). Our approach consists of cleaning the data, selecting informative features, handling class imbalance and training regression models to classify individuals as healthy (-1) or affected (+1).

II. Data Preparation

A. Dataset Description

The BRFSS is an annual health survey conducted in the United States. Its questionnaire contains core questions and optional modules covering health conditions, lifestyle and demographic characteristics.

Our dataset contains 328,135 observations and 321 features. A major challenge is the large number of missing values, including unanswered questions and responses such as "Don't know" or "Refused". Furthermore, the dataset is highly imbalanced, with substantially fewer positive than negative observations.

B. Data Cleaning

We first converted potential missing-value codes (7, 77, etc. and 9, 99, etc.) into NaN values. Features were classified as categorical or continuous according to their number of unique values, using a threshold of 10.

Missing values were replaced by the mode for categorical features and the median for continuous features. We also introduced binary missingness indicators, since the absence of an answer may itself contain predictive information. For example, some questions may only be asked to individuals with particular health conditions.

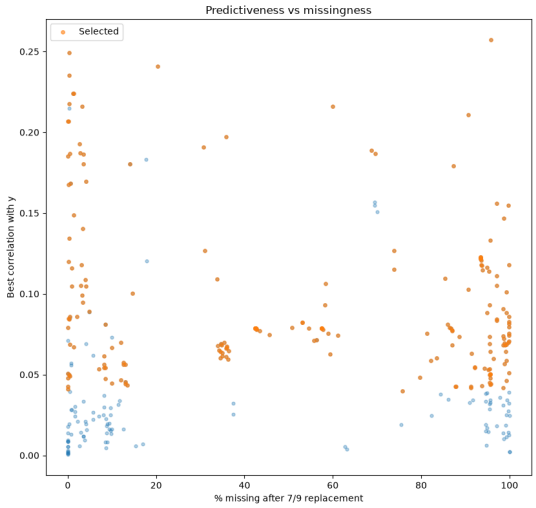
Finally, features were standardized using the training-set mean and standard deviation. The same transformations were applied to validation and test data to avoid data leakage.

C. Feature Selection

To reduce dimensionality, we selected up to 200 features based on their absolute Pearson correlation with the target. For each feature, we considered both its observed values and its missingness indicator, retaining the larger correlation score.

We required at least 40 valid observation pairs to compute a correlation and at least 30 observed values to evaluate a feature's numerical information. These constraints reduce unreliable correlations caused by insufficient data.

Additionally, features with mutual absolute correlation above 0.98 were considered redundant, and only one was retained. Unlike our initial approach, we did not systematically discard features with more than 95% missing values, since missingness can be informative.



III. Method

A. Evaluation Metric

Due to the class imbalance, accuracy alone is not a reliable evaluation metric: predicting every individual as healthy could produce high accuracy while failing to detect any affected individuals.

We therefore prioritized the F1 score, defined as

$$F_1 = 2\frac{P \cdot R}{P + R}, \tag{1}$$

where $P = TP/(TP + FP)$ is precision and $R = TP/(TP+FN)$ is recall. The F1 score balances the detection of positive cases against false positive predictions.

B. Cross-Validation and Class Balancing

We implemented stratified 5-fold cross-validation. The dataset was divided into five subsets preserving approximately the original class proportions. Each fold was used once for validation while the remaining four were used for training.

To address class imbalance, we undersampled the majority class in the training data to obtain approximately 90% healthy and 10% affected individuals. Validation data retained the original distribution to provide a realistic performance estimate.

Feature selection and preprocessing were performed separately within each training fold to avoid information leakage.

C. Regression Methods

We explored ridge regression and regularized logistic regression, as introduced in the lectures.

Our best-performing approach was ridge regression, which minimizes the squared prediction error with an L_2 regularization term:

$$\min_w \frac{1}{2N}\|y - Xw\|_2^2 + \lambda\|w\|_2^2. \tag{2}$$

Although ridge regression produces continuous predictions, these can be converted into binary labels using a classification threshold.

D. Hyperparameter Selection

The main hyperparameters investigated were the regularization strength λ, classification threshold, number of selected features, correlation requirements and training class proportions.

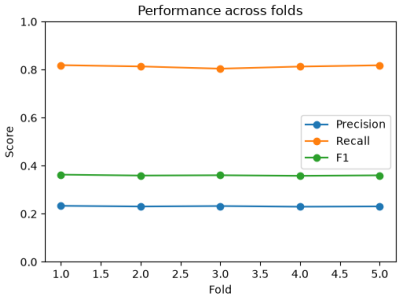
Our final configuration used $\lambda = 0.01$, a classification threshold of -0.6, up to 200 selected features and an 90/10 training class distribution.

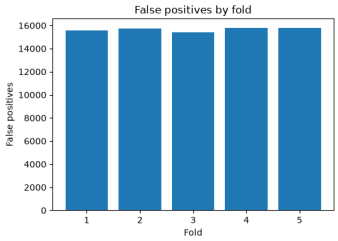
Lowering the classification threshold increases sensitivity to positive cases, potentially improving recall at the cost of more false positives. The regularization parameter controls model complexity and helps limit overfitting.

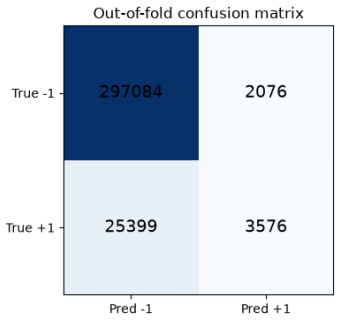
IV. Discussion

Our final submission achieved an F1 score of 0.415 and an accuracy of 0.835 on AIcrowd. These results illustrate the difficulty of predicting coronary heart disease from an imbalanced questionnaire dataset.

Cross-validation provided a more reliable estimate of model performance, although our preliminary comparisons suggested limited differences compared with a single validation split. Its main advantage remains reducing dependence on one particular data split.







With more time, we would perform a more systematic hyperparameter search, particularly for the classification threshold, regularization strength and class-balancing ratio. We would also investigate alternative feature selection strategies and more carefully interpret missing-value codes using the BRFSS documentation.

Overall, our results show that careful preprocessing, informative feature selection and appropriate evaluation metrics are essential for classification on imbalanced health datasets.

V. References

1) CDC, "2015 BRFSS Overview." https://www.cdc.gov/brfss/annual_data/2015/pdf/overview_2015.pdf
2) CDC, "BRFSS Weighting Methodology." https://www.cdc.gov/brfss/annual_data/2015/pdf/weighting_the-data_webpage_content.pdf
3) CDC, "2015 BRFSS Calculated Variables." https://www.cdc.gov/brfss/annual_data/2015/Summary_Matrix_15_version12.html
4) CDC, "2015 BRFSS Codebook." https://www.cdc.gov/brfss/annual_data/2015/pdf/codebook15_llcp.pdf